Mohammad Akram

Senior Full Stack Engineer | AI Agents | Angular | React | Next.js | Node.js

Lucknow, India | +91 9026468433 | aakram1708@gmail.com
Portfolio: mohammad-akram.onrender.com | GitHub: github.com/erakram

10+ Years	3 Live AI Apps	Angular / React	Node.js
Full stack engineering	Multi-agent, RAG, tool use	Next.js, TypeScript, Vue.js	REST APIs, MongoDB

PROFESSIONAL SUMMARY

Senior Full Stack Engineer with 10+ years of experience building scalable web and enterprise applications with Angular, React, Next.js, TypeScript, JavaScript, Node.js and Vue.js. Recently built AI agents and LLM applications, including a multi-agent healthcare revenue-cycle platform, with Python, the Claude API (Anthropic), FastAPI, LLM function calling, Retrieval-Augmented Generation (RAG) and hybrid search. Delivered AI-powered and multi-cloud lab platforms for investment banking, ESG reporting and cloud training, and uses AI coding agents (Claude Code, OpenAI Codex) and custom sub-agents to speed up delivery.

CORE SKILLS

Frontend	Angular (9-21), React, Next.js, TypeScript, JavaScript (ES6+), Vue.js, Nuxt.js, jQuery, HTML5, CSS3, NX Monorepo, Microsoft Teams Apps
Backend	Node.js, FastAPI, REST APIs, SQLAlchemy, Alembic, Server-Sent Events (SSE), JWT Auth
AI / LLM	AI Agents, Claude API (Anthropic), LLM Function Calling / Tool Use, Retrieval-Augmented Generation (RAG), Agentic RAG, Embeddings, Vector Search, Hybrid Search (BM25 + Semantic), Web Search & Fetch Tools, Structured Outputs (JSON Schema), Prompt Engineering, Human-in-the-Loop, LLM Evals, PHI Masking, Prompt-Injection Defence, Groq
AI Dev Tools	Claude Code, OpenAI Codex, Agentic Coding, AI-Assisted Development, Custom AI Coding Agents (Sub-agents), Multi-Agent Workflows
Healthcare	Revenue Cycle Management, X12 EDI (837P, 835, 276/277), FHIR (Da Vinci PAS), ICD-10-CM, CPT, E/M Coding, CARC/RARC Denial Codes, Prior Authorization
Databases	PostgreSQL, MongoDB, SQLite, SQLite FTS5 (Full-Text Search)
Testing & Quality	pytest, Unit & Integration Testing, Structured Logging, Observability, Caching, Retries & Timeouts
Practices	Agile/Scrum, System Design, API Integration, Performance Optimization, Debugging, Code Review, Cross-Functional Collaboration

PROFESSIONAL EXPERIENCE

Senior Software Engineer
Nov 2020 - Present

Spektra Systems, Bengaluru (on payroll of IMAClouds, Lucknow)

Spektra Systems **Angular 21, NX Monorepo, TypeScript**

- Develop the CloudLabs Admin Portal (Angular 21) and Experience and Learner portals (Angular 9-14, NX).
- Deliver provisioning, access control, reporting and monitoring features for multi-cloud lab environments.
- Build reusable UI components and NX monorepo libraries, integrate REST APIs and work with MongoDB-backed services.

IMAClouds Projects (in parallel) **React, Next.js, Vue.js, Nuxt.js, Angular**

- Built the GreenFi ESG Reporting platform with React and Next.js for ESG data collection and sustainability reporting.
- Developed Nexme AI with Vue.js and Nuxt.js, delivering personalized insights and AI-agent workflows.
- Built Klynt AI (Angular 21), an AI platform automating sell-side M&A deal tasks for investment banks and advisory firms.
- Developed Melville (Angular 14), an AI podcast app for audio upload, transcription, key points and audio generation.

- Use AI coding agents (Claude Code, OpenAI Codex) and build custom sub-agents to delegate and automate development tasks, speeding up delivery.
- Collaborate with product, QA and backend teams in Agile sprints to ship scalable, maintainable, production-ready features.

Senior Software Engineer
May 2018 - Oct 2020

IMAClouds, Lucknow

- Developed full stack web applications for IMAClouds clients using Angular, React, Vue.js, Node.js and JavaScript.

Software Engineer
Sep 2016 - Feb 2018

VLogic Labs, Lucknow

- Developed web applications using Ruby on Rails, Node.js, Angular, Vue.js and JavaScript.

Software Engineer
Mar 2016 - Aug 2016

Akaruilabs India, Lucknow

- Developed application features using Ruby on Rails, Node.js and JavaScript.

Junior Software Engineer
Sep 2015 - Feb 2016

Omninet Technologies, Lucknow

- Developed web applications using .NET and JavaScript.

PERSONAL AI AGENT PROJECTS

HealthOps AI - Multi-Agent Healthcare Operations Platform LIVE

Python, FastAPI, PostgreSQL, Next.js 15, React 19, Groq / Claude API, Docker | Built with Claude Code

https://healthops-ai-2.onrender.com

- Hospital workflow run by specialist AI agents, with a person approving every decision: OPD front desk, claim intake, prior authorization, medical coding, denials and appeals, A/R follow-up, patient billing and bank reconciliation.
- Front desk Intake Agent reads complaints in English, Hinglish and Hindi; red-flag rules the model can never clear; Coding Agent searches 74,719 ICD-10-CM codes with 2021 E/M rules for the visit level.
- Real X12 837P/835/276/277 and FHIR Da Vinci PAS; appeal and prior-auth letters cite only verified evidence.
- Document Agent extracts claim fields with highlight boxes on the PDF; a 17-check rules engine and Claims Review Agent catch billing errors.
- Prior auth: Clinical Criteria Agent checks notes against payer policy and asks the physician for missing details.
- Denial Triage and Appeal Agents read the payer's 835, recommend the next step and draft cited appeal letters.
- A/R follow-up, patient statements and payment plans, and bank deposits matched to 835s by EFT trace number.
- PHI masking on every LLM call, role-based access, hash-chained audit trail and prompt-injection defence.
- 350 pytest tests and per-agent evals: prior auth 12/12, E/M and CPT 100%, red-flag recall 12/12, routing 95%.

Employee Support AI Agent LIVE

Python, FastAPI, Groq LLM, SQLite, Docker | Built with Claude Code

https://employee-support-agent.onrender.com/

- AI agent that answers employee questions (reporting lines, leave balances, HR policies) by calling tools that query a company database instead of guessing.
- Agent loop with LLM function calling, argument validation, parameterized SQL and a step limit; the UI shows which tools produced each answer.
- Privacy by design (salary data is never exposed to tools), TTL/LRU caching, retries with backoff, timeouts, latency logging and 26 pytest tests.

Document Research AI Agent (Agentic RAG) LIVE

Python, FastAPI, Groq LLM, model2vec Embeddings, SQLite FTS5, NumPy, Docker | Built with Claude Code

https://document-research-agent.onrender.com/

- Agentic RAG: users upload PDFs and ask questions; the agent searches, rephrases queries and answers with citations to exact passages and pages.
- Hybrid retrieval: semantic vector search (embeddings) plus BM25 keyword search (SQLite FTS5), merged with Reciprocal Rank Fusion.
- Every citation is validated against retrieved passages, so no source is invented; PDF ingestion with overlapping chunks and 36 pytest tests.

PROJECTS

CloudLabs Admin Portal **Angular 21** Admin platform for cloud labs: users, provisioning, access control and monitoring.

GreenFi ESG Reporting Platform **React, Next.js** ESG data collection and sustainability reporting platform.

Klynt AI **Angular 21** AI platform automating repetitive sell-side M&A deal tasks for investment banks and advisory firms.

CloudLabs Experience Portal **Angular 9/12/14, NX** Virtual labs for demos, product trials, POCs and workshops.

CloudLabs Learner Portal **Angular 9/12/14, NX** Virtual labs, certifications, automated validations and progress tracking.

CloudLabs Practice Test (Web and Microsoft Teams App) **Angular 9-12** Microsoft Cloud certification practice tests.

Melville App **Angular 14** AI podcast show notes: transcripts, key points, keywords and audio generation.

Nexme AI **Vue.js, Nuxt.js** AI platform for career and growth decisions with personalized insights and AI-agent workflows.

Oddup **Ruby on Rails** Startup intelligence and investment analytics platform.

Telephone Installer **Ruby on Rails** Stripe subscriptions and Stripe webhooks.

ScootQ **Vue.js, Node.js, MongoDB** Scan in-store products and find lower prices elsewhere.

Space War **Vue.js, Node.js** Fast-paced strategic two-player card game.

Cosmo Wards **jQuery, MongoDB** Consumer and merchant food ordering apps with promos and coupons.

EDUCATION

Master of Computer Applications (MCA) Shri Ramswaroop Memorial College of Engineering and Management (UPTU), Lucknow **2012 - 2015**

Bachelor of Science (B.Sc.) Lucknow University, Lucknow **2010**

ADDITIONAL INFORMATION

Languages: English, Hindi, Urdu | Notice Period: 30 Days